

GPU-Heat vs. Nsight Compute: Supplementary Attribution Comparison

1 Methodology

We compare GPU-Heat’s attribution (Attention rollout and SHAP) against Nsight Compute’s per-instruction stall sampling on the same three convolution-kernel variants used in the submission. For each kernel, we run inference across all three fine-tuning phases and select the phase whose predicted slowdown is highest for the attribution comparison, reflecting the worst-case assumption required in a real-time context; results from the other phases are not included here.

Ground truth (GT) from Nsight. Nsight’s stall sampler records stalls at the *waiting* instruction rather than the *causing* one. We build a composite ground-truth relevance score per normalized PTX instruction from three stall counters given by Nsight Compute:

$$s_{GT} = 0.60 \frac{\text{stall_lg}}{\sum \text{stall_lg}} + 0.30 \frac{\text{stall_mio}}{\sum \text{stall_mio}} + 0.10 \frac{\text{stall_long_sb}}{\sum \text{stall_long_sb}} \quad (1)$$

- **stall_lg**: memory-instruction queue full : most direct signal of memory-system pressure (weight 0.60).
- **stall_mio**: MIO pipeline queue full : one causal step removed (weight 0.30).
- **stall_long_sb**: waiting on an in-flight memory result : lands on the consumer, not the cause (weight 0.10).

Scores are normalized to $[0, 1]$; instructions with $s_{GT} < 0.05$ are treated as non-relevant.

Metrics (cutoff $k = 10$). **Precision@10**: fraction of the top-10 attributed instructions that are GT-relevant. **NDCG@10**: rewards ranking the most GT-relevant instructions near the top. **Hit@1**: whether Nsight’s single highest-scoring instruction appears anywhere in the top-10.

2 Results

2.1 Kernel 1 : Global Memory Only

Slowdown: $2.28\times$ (predicted $2.35\times$ – $2.23\times$ across phases). Both signals recover the dominant **image** and **mask** global loads; Attention’s top hit is the mask pointer-advance **add.s64**, SHAP’s top hit is a mask load.

Table 1: Attribution metrics, Kernel 1.

Signal	NDCG	Precision@10	Hit@1
Attention	0.62	0.60	
SHAP	0.44	0.40	

2.2 Kernel 2 : Image Tiled in Shared Memory

Slowdown: $2.29\times$ (best-phase prediction $2.28\times$, $< 0.4\%$ error). Attention concentrates on the **shared-memory tile-fill** loads/stores; SHAP’s top hit is the remaining **global mask load**, which Nsight ranks as secondary, hence SHAP misses Hit@1.

Table 2: Attribution metrics, Kernel 2.

Signal	NDCG	Precision@10	Hit@1
Attention	0.35	0.90	
SHAP	0.23	0.60	$\times$

2.3 Kernel 3 : Image and Mask Both Tiled

Slowdown: $2.29\times$ (best-phase prediction $2.31\times$). Both signals converge on the **shared-memory tile-fill path** for the mask; this is the strongest three-way agreement with Nsight across all variants.

Table 3: Attribution metrics, Kernel 3.

Signal	NDCG	Precision@10	Hit@1
Attention	0.61	0.60	
SHAP	0.43	0.20	

3 Summary

Across all three kernels, both attribution signals achieve Hit@1 in 5/6 cases and Precision@10 between 0.20 and 0.90. Nsight systematically attributes stalls to the dependent (downstream) instruction rather than the causing memory access; GPU-Heat’s attribution consistently recovers both the causing instruction and its dependency chain, which we interpret as a more complete localization signal for contention-sensitive code regions than isolation-based stall sampling alone.